Software Requirements Specification

Inventory and Sales Tracker for Small Shops

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1. Introduction

Purpose: This document specifies the requirements of the Inventory and Sales Tracker, a web-based application for small retail shops, so that developers, testers, and stakeholders share one understanding of what will be built.

Scope: Many small shops use paper registers or memory to track stock and sales, which causes stock-outs, billing mistakes, unrecorded credit (udhaar), and no clear view of profit. The system digitizes this work through a frontend, a backend, and a database: products and stock, supplier purchases, sales with printable receipts, customer credit, and reports. Online payments, e-commerce, and multi-branch support are outside version 1.0.

Definitions: POS (point of sale: the billing screen), Udhaar (credit given to a customer who pays later), Reorder level (minimum stock before restocking), CRUD (create, read, update, delete), JWT (token used for secure login). Format follows IEEE Std 830-1998.

2. Overall Description

Product perspective: A new, self-contained web application with three tiers: a React frontend in the browser, a Node.js/Express REST API for business logic, and a PostgreSQL database. It runs in any modern browser on a computer, tablet, or phone.

Product functions: role-based login; product and category management; supplier and purchase records; billing with automatic stock update and receipts; customer credit tracking; low-stock alerts; and daily, weekly, and monthly reports with charts.

User	Description and permissions
Owner	Full access: manages users, products, prices, suppliers, purchases, customers, and all reports including profit.
Cashier	Limited access: creates sales, views product availability, records customer payments. Cannot change prices, delete records, or view profit.

Constraints and assumptions: The system must be web-based with a relational database; passwords are stored only as salted hashes; stock-changing operations run inside database transactions. It is assumed that the shop has a device with a browser, a stable network, and (optionally) a printer, and that the owner enters product data at setup.

3. Specific Requirements

3.1 Functional Requirements (H = High, M = Medium, L = Low)

ID	Requirement	Priority
FR-1	Users shall log in with a username and password, and features shall be restricted by role (Owner or Cashier).	H
FR-2	The owner shall add, edit, and deactivate user accounts.	M
FR-3	The system shall add, edit, and delete products (name, category, cost price, sale price, quantity, reorder level) and manage categories.	H
FR-4	The system shall support searching and filtering products by name or category.	M
FR-5	The system shall alert for every product at or below its reorder level.	H
FR-6	The system shall manage suppliers and record purchases (date, supplier, items, total), increasing stock accordingly.	H
FR-7	A billing screen shall let users add items by search or selection with automatic totals.	H
FR-8	Saving a sale shall reduce stock automatically and reject quantities above available stock.	H
FR-9	The system shall support full, partial, and credit payment; unpaid amounts are added to the customer balance.	H
FR-10	The system shall generate a printable receipt for each sale; the owner may void a sale, restoring stock and balance.	M

ID	Requirement	Priority
FR-11	The system shall manage customers, record their payments, and show outstanding balances and history.	H
FR-12	The system shall generate daily, weekly, and monthly sales reports and calculate profit (sale price minus cost price) for the owner.	H
FR-13	The system shall show charts of sales trends and best-selling products, and allow export to PDF or CSV.	L

3.2 External Interface Requirements

- User interface: responsive web pages (dashboard, products, billing, customers, reports) with input validation and clear error messages.
- Hardware: no special hardware; optional receipt printer and keyboard-type barcode scanner.
- Software and communication: REST API with JSON over HTTP/HTTPS; JWT sent in the request header; PostgreSQL accessed by the backend only.

3.3 Non-Functional Requirements

Attribute	Requirement
Performance	Pages load within 3 seconds and a sale is saved within 2 seconds for up to 10 concurrent users.
Security	Hashed passwords, JWT sessions with expiry, server-side role checks, protection against SQL injection and XSS.
Reliability	Sales and stock updates are atomic; daily database backups; 99% availability during shop hours.
Usability	A new cashier can create a bill after 30 minutes of training; simple, consistent interface.
Maintainability / Portability	Modular layered code; runs on any OS with a modern browser (client) and Node.js with PostgreSQL (server).

4. Data Requirements

Entity	Key attributes	Relationships
users, categories	id, name, username, role, password_hash / id, name	Users record sales and purchases
products	id, name, category_id, cost_price, sale_price, stock_qty, reorder_level	Belongs to a category
suppliers, purchases	id, name, phone / id, supplier_id, date, total (+ purchase_items)	Supplier has many purchases
customers, payments	id, name, phone, balance / id, customer_id, amount, date	Customer has many sales and payments
sales, sale_items	id, customer_id, user_id, date, total, paid_amount / sale_id, product_id, qty, unit_price, unit_cost	Sale has many items

Business rules: stock can never be negative; cost price is stored with each sale item so past profit stays correct; customer balance = credit sales minus payments; products used in past sales are deactivated, not deleted.

5. Key Use Case: Create a Sale

Actor: Cashier or Owner. **Precondition:** logged in; products have stock. **Postcondition:** sale saved, stock reduced, receipt available, any credit added to the customer balance.

1. The cashier opens the billing screen and adds products with quantities; totals update automatically.
2. The cashier optionally selects a customer and enters the amount paid; the system shows change or the credit amount.
3. On confirmation the system saves the sale in a transaction, reduces stock, and shows a printable receipt and any low-stock alerts.

Alternate flow: if a quantity exceeds stock the system shows an error; if the payment is partial, a customer must be selected.